

Sacha Arseneault

sacha.arseneault@gmail.com | linkedin.com/in/sacha-ars | github.com/xsachax | sachaa.dev

EDUCATION

University of Ottawa <i>BASc in Software Engineering, Co-op</i>	Ottawa, ON <i>GPA: 3.9/4.0 Expected: Dec 2026</i>
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EXPERIENCE

Microsoft AI <i>Applied AI/ML Intern — Copilot</i> - Modernized Copilot’s agent stack serving 100M+ monthly users—across harness engineering, response quality, and inference—driving its transition toward longer-running agentic workflows. - Increased first-turn KV cache hit rates on low-traffic GPT models from <u>7% to 90%</u> by introducing dynamic cross-conversation GPU routing, saving up to <u>\$1.4M annually</u> in inference costs. - Architected an end-to-end evaluation platform for coding-agent fleets—combining evidence bundling, a 7-axis taxonomy, and LLM judging—profiling <u>1,000+ PRs/week</u> across Microsoft products serving 500M+ monthly users. - Influenced the “autopilot” feature roadmap by proposing <i>self-directed continuity</i> to VP leadership, enabling always-running agents to pursue missions autonomously with zero user intervention.	Mountain View, CA <i>June 2026 - Aug 2026</i>
Tesla <i>Software Engineer Intern — Robotaxi</i> - Established early reliability guardrails for Tesla’s robotaxi platform—across automation testing, observability, and rider feedback pipelines—maximizing readiness for its official release in June 2025. - Configured post-merge canaries validating the complete rider journey—from ride hailing to passenger drop-off—by using mobile emulators and simulated car fleets, flagging deployment issues immediately without impacting live rides. - Built an integration test suite validating telemetry emitted from 6 different sources during real robotaxi rides, reducing full-system verification from <u>15 to 2 minutes</u> and saving <u>150 hrs/year</u> of manual testing work per engineer. (Go, Docker, Kafka) - Implemented fault-tolerant telemetry parsing that recovers <u>100% of valid fields</u> in legacy payloads, preserving accountability and observability for robotaxis running on older Model Y firmware. - Conducted live QA across 200+ physical rides, identifying infotainment and rider experience issues and coordinating resolution with cross-functional teams ahead of Early Access.	Palo Alto, CA <i>May 2025 - Aug 2025</i>
Shopify <i>Software Engineer Intern — Stores</i> - Strengthened Shopify’s store-creation pipeline handling 2M+ annual store creations—with data consistency safeguards and automated recovery—preventing failed or inconsistent creations from disrupting merchant launches. - Protected <u>20,000+ new stores annually</u> from customer-facing data inconsistencies by resolving a 1-in-100 race condition that created corrupted database records during merchant onboarding. - Eliminated the need for on-call engineers to remediate backend store-creation failures by automating recovery across all failure states—including timeouts and platform outages—resolving <u>100% of stuck requests</u>. (Ruby, Rails, GraphQL, MySQL)	Ottawa, ON <i>Jan 2025 - Apr 2025</i>
Motorola <i>Software Engineer Intern — Emergency Call Handling</i> Ciena <i>Software Engineer Intern — Developer Tools & Experience</i>	Gatineau, QC Sep 2024 - Dec 2024 Ottawa, ON Jan 2024 - Apr 2024

LEADERSHIP

IEEE uOttawa - Webmaster - Led a team of 10 developers in building 4 open-source web-apps driving <u>5,000+ visits</u> and <u>40,000+ impressions</u> per month, serving a community of over 30,000 engineering students across Eastern Canada. - Hosted 2 workshops (Git/React) and 3 review sessions (Data Structures and Algorithms) with <u>75+ attendees each</u>.	2022 - 2026
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PROJECTS

Concurrency Debugger for Go <i>Golang, Websocket, GinkGo</i> A cross-platform Go debugger that visualizes goroutines, channels, and synchronization through low-level Linux/macOS backends, exposing execution control and runtime inspection to IDEs and coding agents via DAP.	2026
Eco Rewards App <i>Remix, TypeScript, Express.js, PostgreSQL, ShadCn, Auth0</i> - A full-stack points-tracking app rewarding users for sustainable day-to-day actions, with AI receipt parsing. Won 1st Place out of <u>842</u> participants in a top sponsor track at Hack the North, Canada’s largest hackathon.	Waterloo, ON 2024

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, Java, SQL, Ruby Frameworks/Libraries: Express, Next, React, Redux, PostgreSQL, Rails, Supabase, FastAPI, Scikit-learn, Pandas, Torch Tools: Git, Docker, Linux, Bash, Prometheus, Grafana, OpenTelemetry, Kafka, Redis, GraphQL, gRPC, Protobuf	
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